that is,

$$y^2 - 8y - 65 = 0.$$

Solving this we get solutions $y = 13$ and $y = -5$. Substituting this we get two quadratic equations

$$x^2 - 5x - 13 = 0 \text{ and } x^2 - 5x + 5 = 0$$

which can be solved by usual techniques. The solutions obtained for these two equations together give solutions as $\frac{5 \pm \sqrt{77}}{2}, \frac{5 \pm \sqrt{5}}{2}$.

Example 3.24

Solve the equation $(2x-3)(6x-1)(3x-2)(x-2) - 5 = 0$.

Solution

The given equation is same as

$$(2x-3)(3x-2)(6x-1)(x-2) - 5 = 0.$$

After a computation, the above equation becomes

$$(6x^2 - 13x + 6)(6x^2 - 13x + 2) - 5 = 0.$$

By taking $y = 6x^2 - 13x$, the above equation becomes,

$$(y+6)(y+2) - 5 = 0$$

which is same as

$$y^2 + 8y + 7 = 0.$$

Solving this equation, we get $y = -1$ and $y = -7$.

Substituting the values of y in $y = 6x^2 - 13x$, we get

$$6x^2 - 13x + 1 = 0$$

$$6x^2 - 13x + 7 = 0$$

Solving these two equations, we get

$$x = 1, x = \frac{7}{6}, x = \frac{13 + \sqrt{145}}{12} \text{ and } x = \frac{13 - \sqrt{145}}{12}$$

as the roots of the given equation.

EXERCISE 3.4

1. Solve : (i) $(x-5)(x-7)(x+6)(x+4) = 504$ (ii) $(x-4)(x-7)(x-2)(x+1) = 16$
2. Solve : $(2x-1)(x+3)(x-2)(2x+3) + 20 = 0$

3.8 Polynomial Equations with no Additional Information

3.8.1 Rational Root Theorem

We can find a few roots of some polynomial equations by trial and error method. For instance, we consider the equation

$$4x^3 - 8x^2 - x + 2 = 0 \quad \ldots (1)$$

This is a third degree equation which cannot be solved by any method so far we discussed in this chapter. If we denote the polynomial in (1) as $P(x)$, then we see that $P(2) = 0$ which says that $x - 2$ is a factor. As the rest of the problem of solving the equation is easy, we leave it as an exercise.

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